A Short History of Computing

The history of computing stretches back long before the electronic computer. Charles Babbage designed the Analytical Engine in 1837, a mechanical machine that is widely regarded as the first design for a general-purpose computer. Ada Lovelace wrote the first algorithm intended for the Analytical Engine, and she is often called the first computer programmer. Her notes described how a machine could manipulate symbols according to rules.

In 1936, Alan Turing introduced the Turing machine, an abstract model of computation that defines what an algorithm can compute. The Turing machine remains the foundation of theoretical computer science. During the Second World War, Turing also helped break German ciphers at Bletchley Park in England. Colossus was one of the first programmable electronic digital computers and was built by British engineers in 1943.

The ENIAC was completed in 1945 at the University of Pennsylvania. It used about 18,000 vacuum tubes and weighed roughly 30 tons. Vacuum tubes were later replaced by transistors, which are smaller, faster and far more reliable. The transistor was invented at Bell Laboratories in 1947 by John Bardeen, Walter Brattain and William Shockley.

Integrated circuits placed many transistors on a single silicon chip. Jack Kilby demonstrated the first integrated circuit in 1958. Gordon Moore observed that the number of transistors on a chip doubles roughly every two years, an observation known as Moore's Law. The microprocessor, a complete processor on one chip, appeared in 1971 with the Intel 4004.

Personal computers brought computing to homes and schools during the 1980s. The World Wide Web was proposed by Tim Berners-Lee at CERN in 1989. The web allowed hypertext documents to be shared across the internet using the HTTP protocol. Machine learning is a branch of artificial intelligence that allows computers to learn patterns from data. Modern machine learning relies on large datasets and powerful graphics processors.